

Wei Wang

Ph.D. Candidate in Mathematics, Peking University
School of Mathematical Sciences | gjmtamag@gmail.com

Education

- **Peking University, School of Mathematical Sciences** Sept. 2022 – Present
Ph.D. Candidate in Mathematics (Advisor: Prof. Zhifei Zhang) Beijing, China
- **Zhejiang University, School of Mathematical Sciences** Sept. 2018 – June 2022
B.S. in Mathematics and Applied Mathematics (Advisor: Prof. Ting Zhang) Hangzhou, China
Thesis: *Homogenization Problem of Second Order Elliptic Systems with Lower Order Terms*

Research Visits and Programs

- **University of Verona, Department of Computer Science** Mar. – Apr. 2026
Visiting Student (Host: Prof. Giacomo Canevari) Verona, Italy
- **HKUST, Department of Mathematics** June – Sept. 2021
Summer Research Program (Advisor: Prof. Tianling Jin) Hong Kong, China

Research Interests

- **Research areas:** Geometric variational problems, elliptic equations, parabolic equations
- **Selected topics:** Homogenization theory, semilinear elliptic and parabolic equations, harmonic maps, liquid crystal theory, nonlocal Laplacian operators

Publications

3. Rupert L. Frank, Tianling Jin, and Wei Wang, On the sharp constants in the regional fractional Sobolev inequalities, *Partial Differential Equations and Applications*, 6 (2025), 1–20.
2. Wei Wang, Uniform estimates of resolvents in the homogenization theory of elliptic systems, *Journal of Differential Equations*, 370 (2023), 1–65.
1. Wei Wang and Ting Zhang, Homogenization theory of elliptic systems with lower order terms in dimension two, *Communications on Pure and Applied Analysis*, 22 (2023), 787–824.

Preprints and Submitted Manuscripts

9. G. Canevari, Haotong Fu, and Wei Wang, Asymptotics of minimizers for Ginzburg–Landau-type functionals in high dimensions, *arXiv:2605.04442*, submitted.
8. Haotong Fu, Huaijie Wang, and Wei Wang, Smooth convergence of Landau–de Gennes minimizers in the Lyuksyutov regime, submitted.
7. Haotong Fu, Huaijie Wang, and Wei Wang, Uniform estimates of Landau–de Gennes minimizers in the vanishing elasticity limit with line defects, *arXiv:2508.01811*, accepted by *SIAM Journal on Mathematical Analysis*.
6. Haotong Fu, Huaijie Wang, and Wei Wang, Improved convergence of Landau–de Gennes minimizers in the vanishing elasticity limit, *arXiv:2507.14955*, submitted.
5. Wei Wang, Wei Wang, and Zhifei Zhang, Existence of weak solutions for two-phase matrix-valued harmonic map flows, *arXiv:2506.16278*, submitted.
Note: the second author Wei Wang (王伟) is a different person.
4. Haotong Fu, Wei Wang, Ke Wu, and Zhifei Zhang, Stratification and rectifiability of harmonic map flows via tangent measures, *arXiv:2504.14880*, submitted.
3. Wei Wang and Zhifei Zhang, Fine structure of rupture set for semilinear elliptic equations with singular nonlinearity, *arXiv:2411.16048*, submitted.
2. Haotong Fu, Wei Wang, and Zhifei Zhang, Quantitative stratification and sharp regularity estimates for supercritical semilinear elliptic equations, *arXiv:2408.06726*, submitted.
1. Wei Wang and Zhifei Zhang, Landau–de Gennes model with sextic potentials: asymptotic behavior of minimizers, *arXiv:2404.00677*, submitted.

Selected Talks

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| • Workshop on Fluid Dynamics Equations, Peking University
<i>Title: Landau–de Gennes Model with Sextic Potentials: Asymptotic Behavior of Minimizers</i> | Jan. 2026 |
| • Jinhua Conference on Harmonic Analysis, Zhejiang Normal University
<i>Title: Stratification for Harmonic Map Flows via Tangent Measures: Rectifiability and Regularity</i> | Nov. 2025 |
| • 3rd National Forum for Doctoral Students in Frontier Mathematics, Peking University
<i>Title: Vanishing Elasticity Limit and Defect Analysis in the Landau–de Gennes Model</i> | Nov. 2025 |
| • Conference on Theory and Computation of Quasicrystals, Nankai University
<i>Title: Improved Convergence of Landau–de Gennes Minimizers in the Vanishing Elasticity Limit</i> | Aug. 2025 |
| • PDE and Geometry Seminar, Zhejiang University
<i>Title: Rectifiability of the Stratification for Harmonic Map Flows via Tangent Measures</i> | June 2025 |

Grants and Projects

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| • Physical Modelling and Mathematical Theory of Liquid Crystals , participant, NSFC, Grant No. 12288101 | 2023 – 2028 |
| • Physical Modeling and Mathematical Theory of Liquid Crystal Systems , participant, National Key R&D Program of China, Grant No. 2023YFA1008801 | 2023 – 2028 |

Teaching Experience

Peking University, Teaching Assistant

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| • Advanced Mathematics (A, B, C), various semesters | 2022 – 2025 |
| • Real Analysis; Functional Analysis (I); Harmonic Analysis | 2023 – 2025 |

Zhejiang University, Teaching Assistant

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| • Mathematical Analysis (I & II) | 2021 – 2022 |
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Honors and Awards

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| • First Prize , Graduate Research Exhibition, School of Mathematical Sciences, Peking University | Apr. 2025 |
| • Third Prize , Graduate Research Exhibition, School of Mathematical Sciences, Peking University | Apr. 2024 |
| • Elite Program for Graduate Students in Mathematics, Peking University | Oct. 2023 |
| • Nomination Award, ICCM Creative Undergraduate Thesis | Aug. 2022 |
| • First Prize , 11th Chinese Mathematics Competitions Finals (CMC) | Apr. 2021 |
| • Nomination Award, S.-T. Yau College Student Mathematics Contest | June 2021 |

Technical Skills

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- **Software:** $\text{\LaTeX}$, MATLAB, Mathematica.
 - **Languages:** Mandarin Chinese (native), English (fluent).